LINEAR ARRAY SP' 26

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1. Introduction

1.1 Overview of Project

Rapid and non-invasive diagnostic technologies are critical for early detection and containment of viral infections. Existing diagnostic methods, such as PCR-based testing, often require specialized equipment, trained personnel, and significant processing time, limiting their effectiveness in real-time screening applications. Breath-based diagnostics present a promising alternative, as exhaled aerosols contain biological particles that can reflect an individual's infection status. Rapid antigen tests provide a good substitute, but still take at least 15-30 minutes for accurate results. Breath-based diagnostics present a promising alternative, as exhaled aerosols contain biological particles that can reflect an individual's infection status.

This project explores the feasibility of developing a device capable of instantly identifying viral infections through analysis of exhaled particles, specifically by leveraging differences in particle mass. While existing systems demonstrate the ability to detect and analyze such particles at larger scales, their size, complexity, and cost restrict widespread deployment.

The primary objective of this work is to establish a proof of concept for scaling down these detection systems to the micro level. In particular, this project focuses on evaluating the manufacturing feasibility of miniaturizing the parallelized linear array detector while maintaining functional performance. By addressing constraints in fabrication, alignment, and material selection, this study aims to determine whether a compact, scalable diagnostic device is achievable.

1.2 Design Requirements and Constraints

The design of the proposed micro-scale detection device must satisfy several key design constraints:

1. Require a 4:1 ratio of electrode width to ion hole diameter
2. We need a high transmittance, which is classified for the purposes of this project as a transmittance level of $\geq$ 60%
3. Electrical connections are necessary between the PCB and electrodes for the charge to be passed through all electrodes
4. Insulation between each of the electrodes to ensure the charge calculated from each electrode is separate
5. Precision hole alignment between electrodes and a robust architecture of the device

2. Documentation

2.1 CAD Assembly Link

2.2 Bill of Materials (BOM)

Article Number	Part Number / Source	Part Name	Quantity Used
1	1388K104 McMaster	Ground Low-Carbon Steel Sheet, 4" x 4", 1/4" Thick	
2	91595A112 McMaster	Dowel Pin, 52100 Alloy Steel, 3mm Diameter, 12mm Long, Packs of 50	
3	91290A117 McMaster	Black-Oxide Alloy Steel Socket Head Screw, M3 x 0.5 mm Thread Size, 12mm Long, Packs of 100	
4	91290A013 McMaster	Black-Oxide Alloy Steel Socket Head Screw, M2 x 0.4mm Thread Size, 6 mm Long, Packs of 100	
5	90591A250 McMaster	Zinc-Plated Steel Hex Nut, Medium-Strength, ISO Class 8, M3 x 0.5 mm Thread Size, Packs of 100	
6	90591A265 McMaster	Zinc-Plated Steel Hex Nut, Medium-Strength, ISO Class 8, M2 x 0.4 mm Thread Size, Packs of 100	
7		PCBs	1
8		Resin	

2.3 Manufacturing Procedures

For spade manufacturing, steel sheet metal should be cut on the Fablight or any other compatible laser cutter. Post cutting, the pieces will need to be processed in a series of ways. Firstly, they must be carefully removed from the sheet metal. Secondly, the sharp edge produced from fabricating the material must be deburred. This can be done with files or a belt sander.

For datum and brackets produced with esd resin, they should ideally be reemed for the critical datum holes. For this, any standard mill should be set up with a hole finder and then reemed with a 3 mm reemer. We designed PLA 3D printed supports to go into the part while milling to support the structure. This specifically only needs to be done on certain holes on the brackets, which is denoted on the part drawings. With ESD resin, there is sometimes finish issue, like bubbles or rought surfaces. For this, we file the pieces down. If there is any holes which have exces resin in them, take a small circular file and slowly file the resin out.

The base brackets made from steel should be manufactured by CNC. For this, the file should be loaded on to the CNC machine and the metal stock properly locked in place. Follow that specific CNC instructions for tooling. Post CNCing, there are specific holes on the part which similarly will needed to be reemed out to get the correct tolerances. These holes are also 3 mm, so use the hole finder and a 3 mm reemer. The exact holes that require reaming are on the part drawing.

2.4 Assembly Procedures

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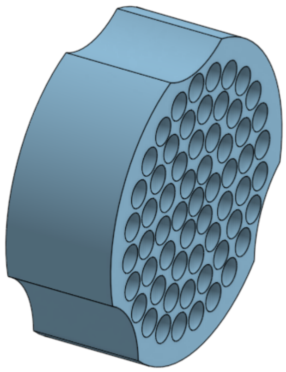
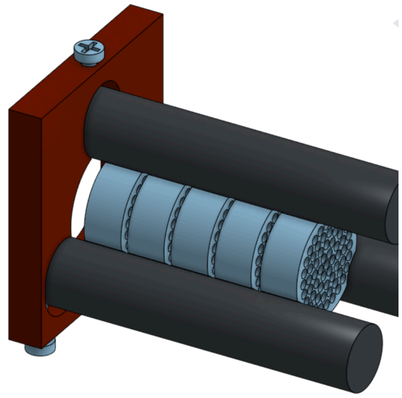
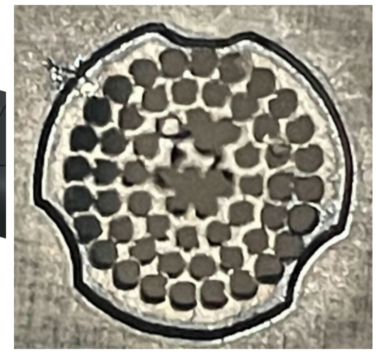
3. Mechanical Analysis & Design Reasoning

3.1 Shape of Electrode and Alignment

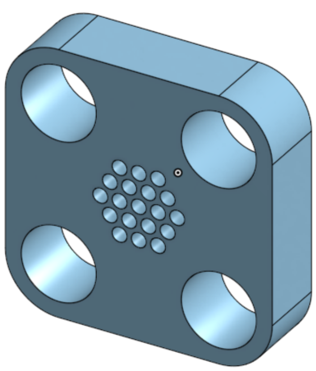
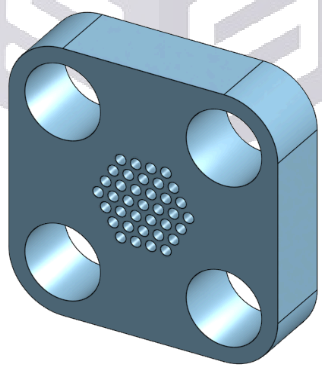
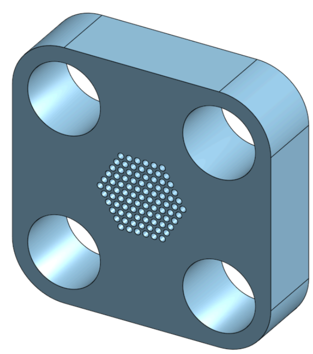
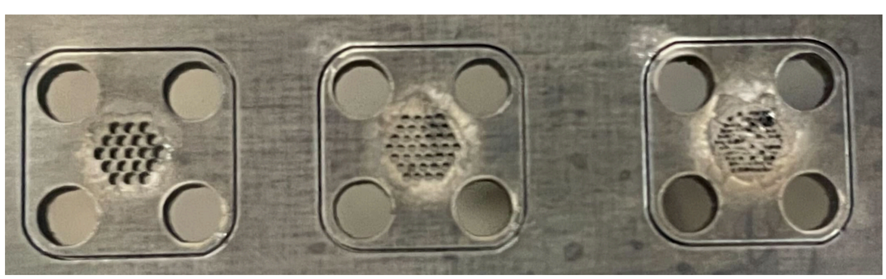
3.1a - Circle Design

Initially, the shape of the electrode would have been held together by 3 rods and compressed by screws on either end. The electrode was designed to be circular-shaped and have 3 negative circular cuts on the edges. This would allow 3 rotational constraints and 2 kinematic constraints. The strength of this design was having a large transmittance. This design wasn't chosen due to the difficulty of assembling, as well as the inability to manufacture on the Fablight due to the close holes. Later, it would be realised that this design would also be difficult to integrate with the PCB without any interface connection.

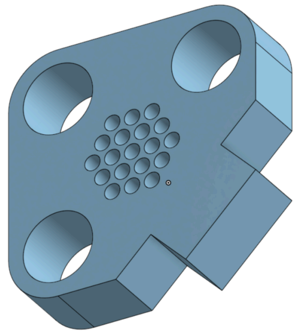
3.1b - Square Design

The second design was a 4- rod system, which had a similar concept of the first (using rods to keep the electrodes in place), however, this design called for the rods to be placed through the electrodes. This would also lock it in 3 rotational, and 2 kinematic directions. 3 different versions of this design (.2mm .4mm .7mm center holes) were designed in order to test manufacturing. The number of holes increased for the smaller sizes, in order to compensate for the lost surface area when shrinking the holes.The main drawback of this design was it only allowed the PCB to interface tangent to the surface when placed on top directly, This meant that the connection would be weak, jeopardizing the signal strength.

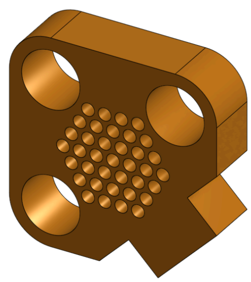





3.1c Spade design

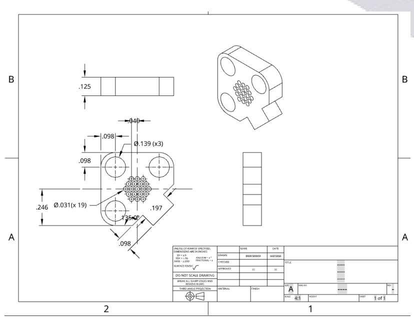
The final iteration of the electrode design was modified to allow the PCB to easily interface with each electrode. The PCB would mount on top of the electrode, resting on the flat face at the “stem”. The center hole diameter was also adjusted slightly, from .700 mm to 0.781 mm, due to the required 4:1 hole ratio, and having ordered 3.125 mm thick metal. This has a total of 19 small holes

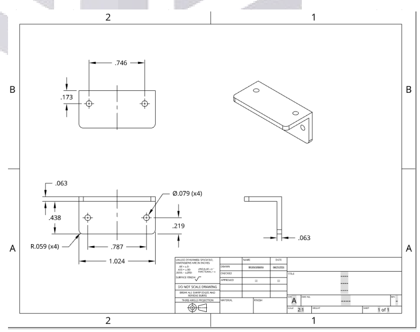


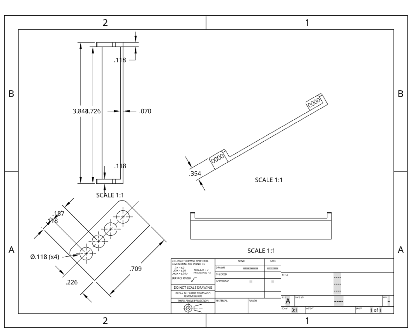
Once the manufacturing process was proven, the final design introduced another ring of center holes, allowing for a higher transmittance. This is a total of 37 small holes.

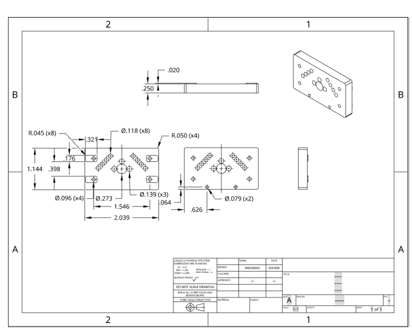


3.1a First Iteration (Rod Locking)









3.2 Materials Analysis

Oxygen-free copper (OFHC) was evaluated as a candidate material due to its exceptional performance in vacuum environments and its superior electrical and thermal conductivity, making it highly suitable for use in electrode components. These properties are particularly advantageous for applications requiring efficient signal transmission and minimal energy loss. However, despite its favorable functional characteristics, OFHC copper presents notable manufacturing challenges at the micro scale. Its high ductility and tendency to undergo strain hardening limit the achievable precision and structural stability during fabrication, thereby constraining its compatibility with conventional machining processes. While advanced fabrication methods such as powder bed fusion could potentially address these limitations, current constraints in equipment availability prevent the use of such techniques in this project.

Stainless steel was also considered as an alternative material, offering moderate electrical and thermal conductivity while providing significantly improved manufacturability. Its compatibility with CNC machining enables high-precision fabrication, which is critical for maintaining tight tolerances in datum features and alignment components such as dowels. As a result, stainless steel is well-suited for structural elements requiring dimensional accuracy and repeatability, and can also be used for electrode applications where extreme conductivity is not the primary requirement.

ESD Resin was used in the creation of alignment brackets, dowels, and spacers. This material was ideal because of its charge control and its mechanical positioning properties. Because of its 3D printing capabilities, custom geometry with high tolerances can be printed, which is ideal for these small alignment features. The ESD filament is good for vacuum systems because they are sensitive to static charge buildup. Where a normal insulating plastic could accumulate charge, which could distribute electric fields, attract particles, and affect ion trajectories, ESD is slightly conductive, so the charge can slowly bleed off instead of building up.

Alumina, specifically 99.7% purity, is an ideal material for a mass spectrometer because of its dimensional stability, vacuum compatibility, and electrical insulation. Alumina is a strong electrical insulator, which allows it to not disrupt the electric fields and ion trajectory. It also has outgassing and doesn't absorb moisture or release volatiles, which is ideal for a vacuum. Alumina has a low thermal expansion coefficient, so it doesn’t shift much with temperature changes.

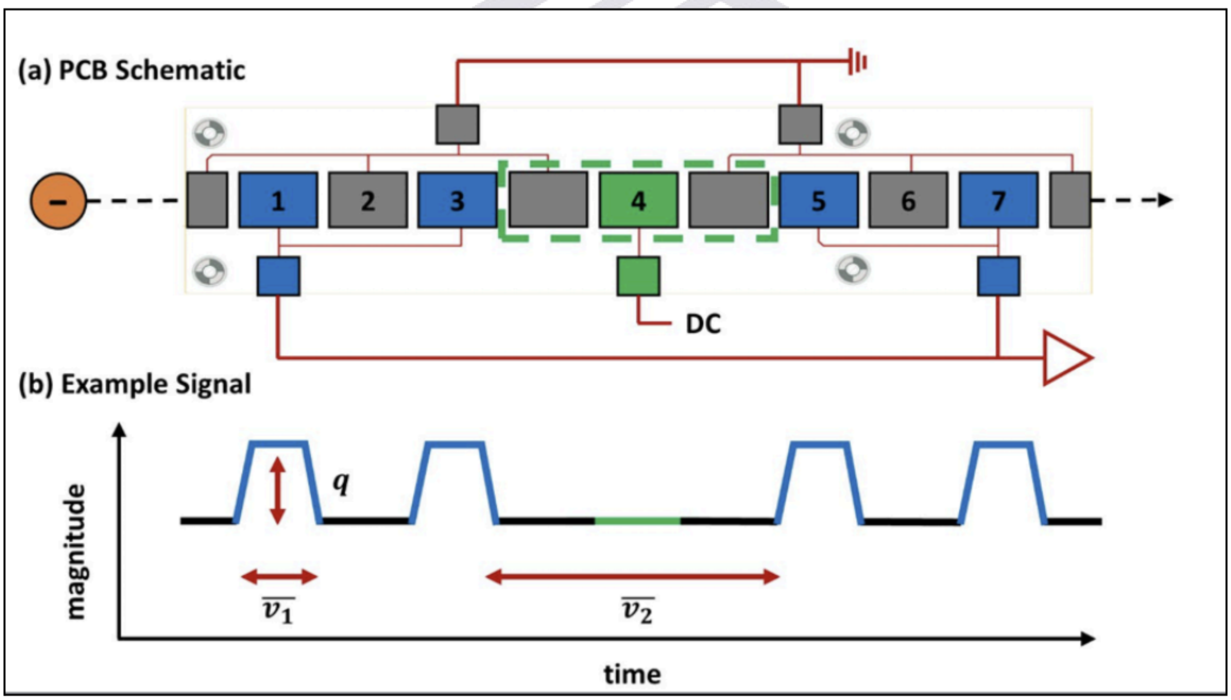
3.3 Manufacturing Methods Research

CNC manufacturing was an ideal choice for manufacturing because of its ability to manufacture parts with high tolerances and precision, better than machining and 3d printing. CNCs work well with metals like stainless steel, providing a high surface quality and durability. The high surface finish makes it ideal for vacuums as well.

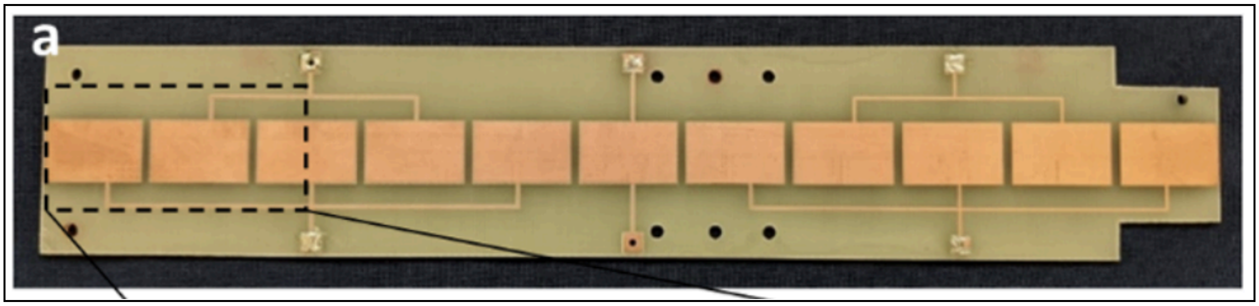
ESD SLA printing is similar to CNC in that it allows for high precision manufacturing, even with small, complex geometries. In our case, it allowed for fast iterations and low cost. When we were working on finalizing and reaming our datum pieces, we were able to manufacture multiple copies and try different reaming methods for a low cost. For reaming the ESD parts to ensure the holes were exactly the correct size, we used a mill and a reaming attachment. ESD resin was also used for the spacers for similar reasons, and because of how malleable the material is, it is easy to file them to be exactly the correct size to perfectly fit the spaces.

Fablight or laser cutting was used to cut steel for the spades because of its precision, speed, and vacuum-compatibility. With thin, flat geometries, laser cutting achieves tight tolerances. For the holes and datum parts of the spades, this was very important. The low cost, fast iteration capabilities worked well for our design as well, allowing us to design multiple different iterations of spades to finalize the best one. One issue with fabricating sheet metal is it produces sharp edges which needed to be deburred and polished.

3.4 Electrical Research



Our research consisted of reading literature on past projects similar to this one and signal behavior. To start off, we observed that BYU and NASA Martian particle detectors simulated and tested various electrode sizes and shapes to start off their process, to provide acceleration to the ion for better reads, a DC amplifier was placed in the middle of the circuit, and we also researched the sizes of ions from virus strains. This initial paper provided us with a basic understanding into how the current system worked.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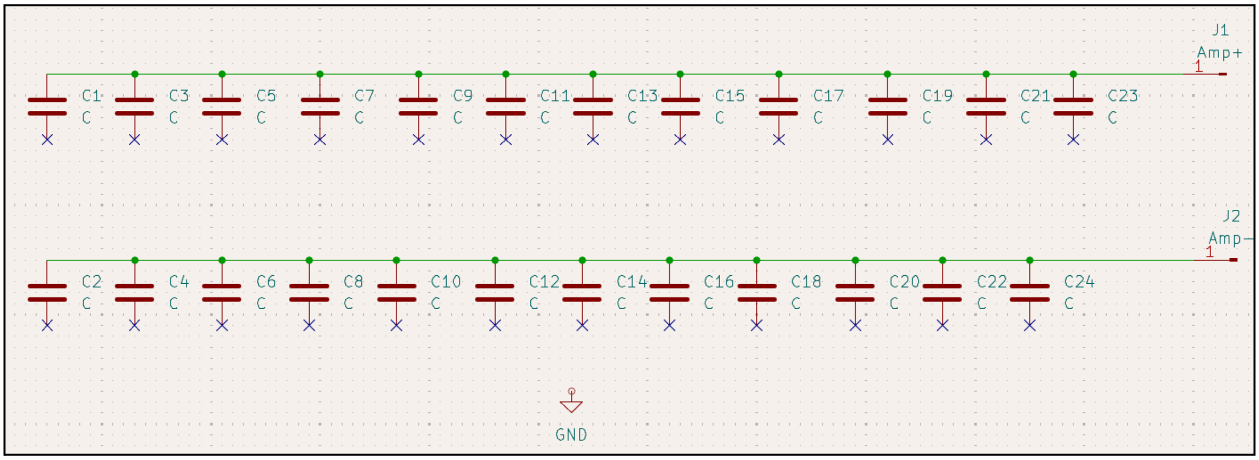
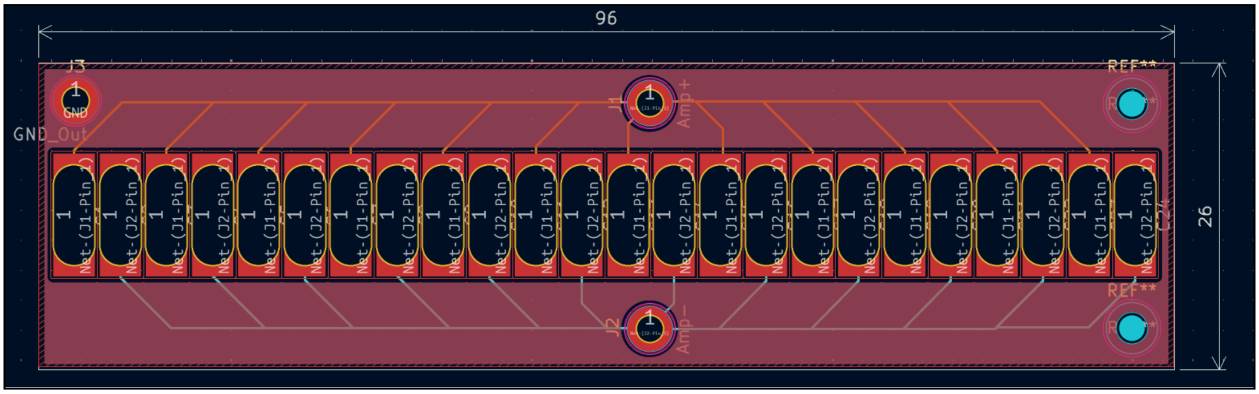
4. Electrical Design

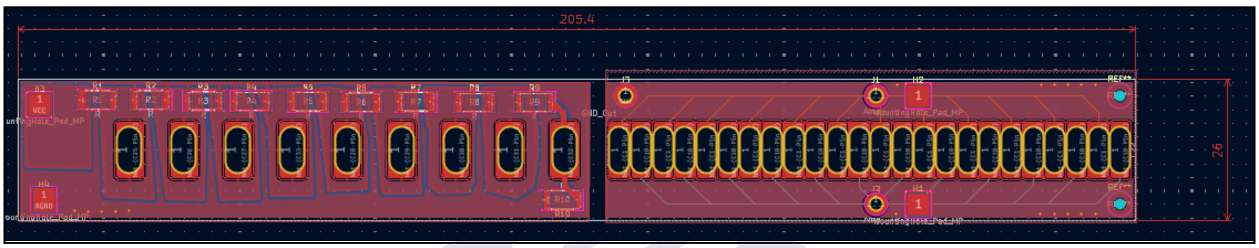
4.1 Schematic

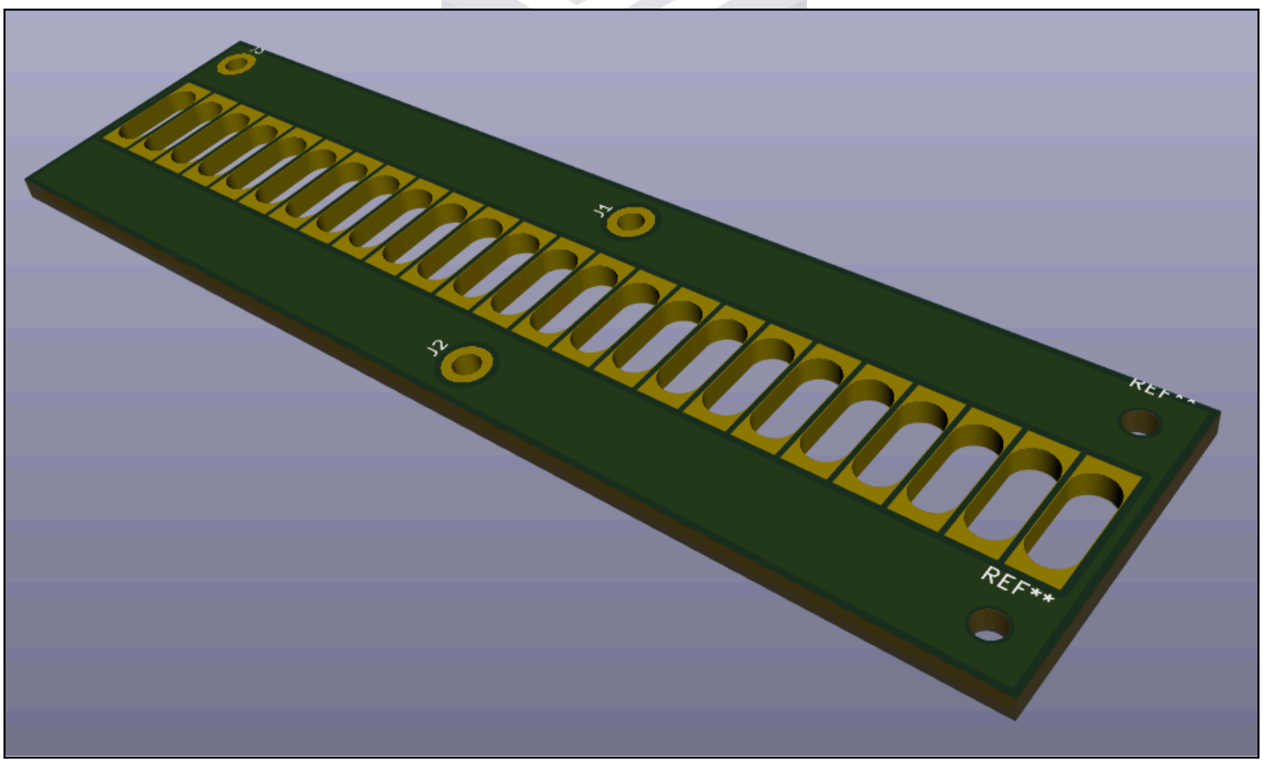


4.2 Footprint

4.2.1 Model 1:

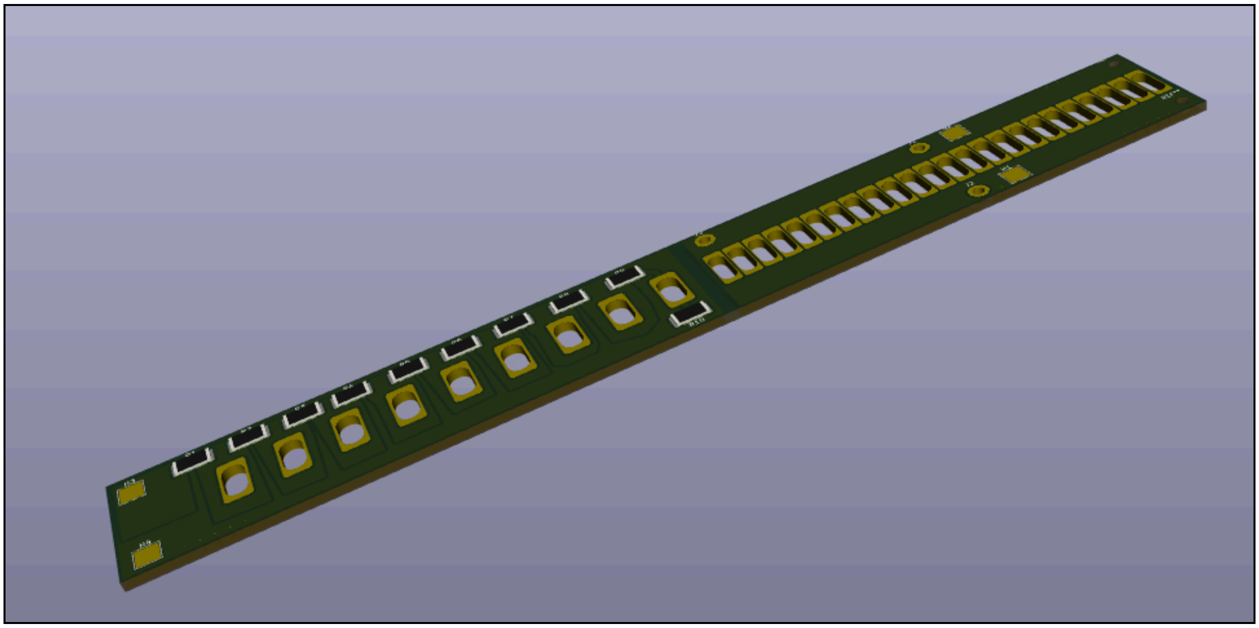






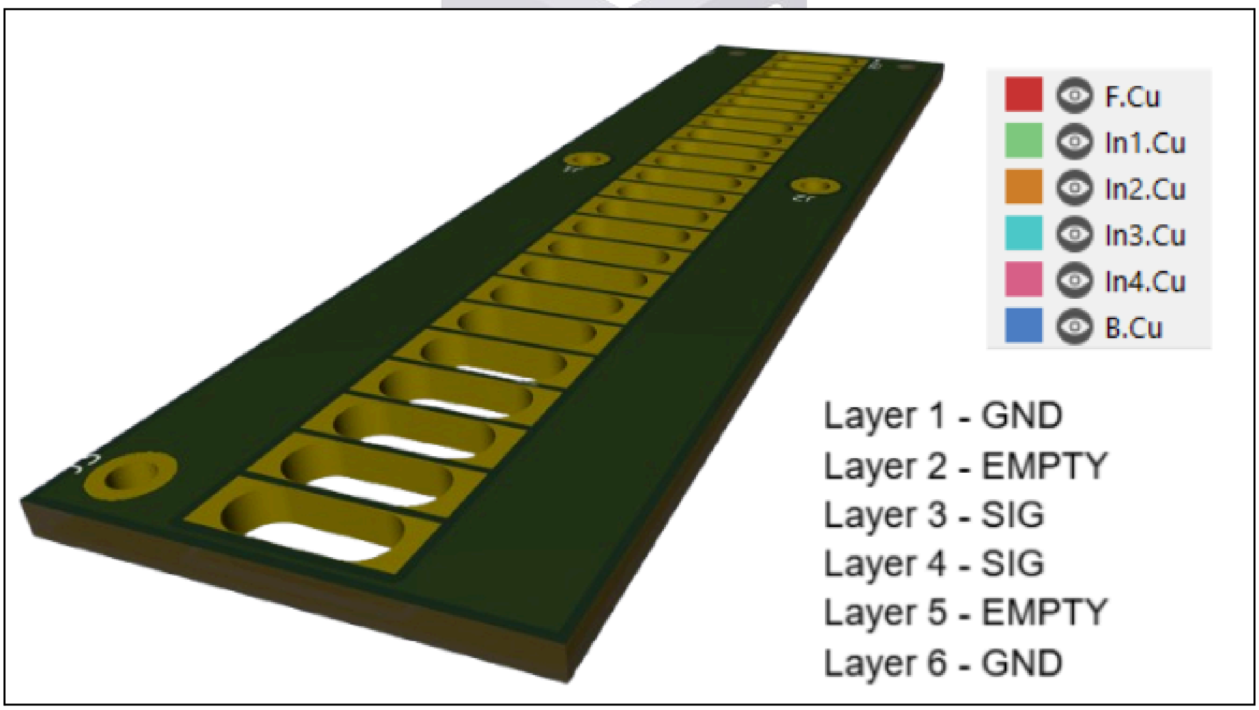
Between each spade there is a distance of 0.6mm.

4.2.2 Model 2:

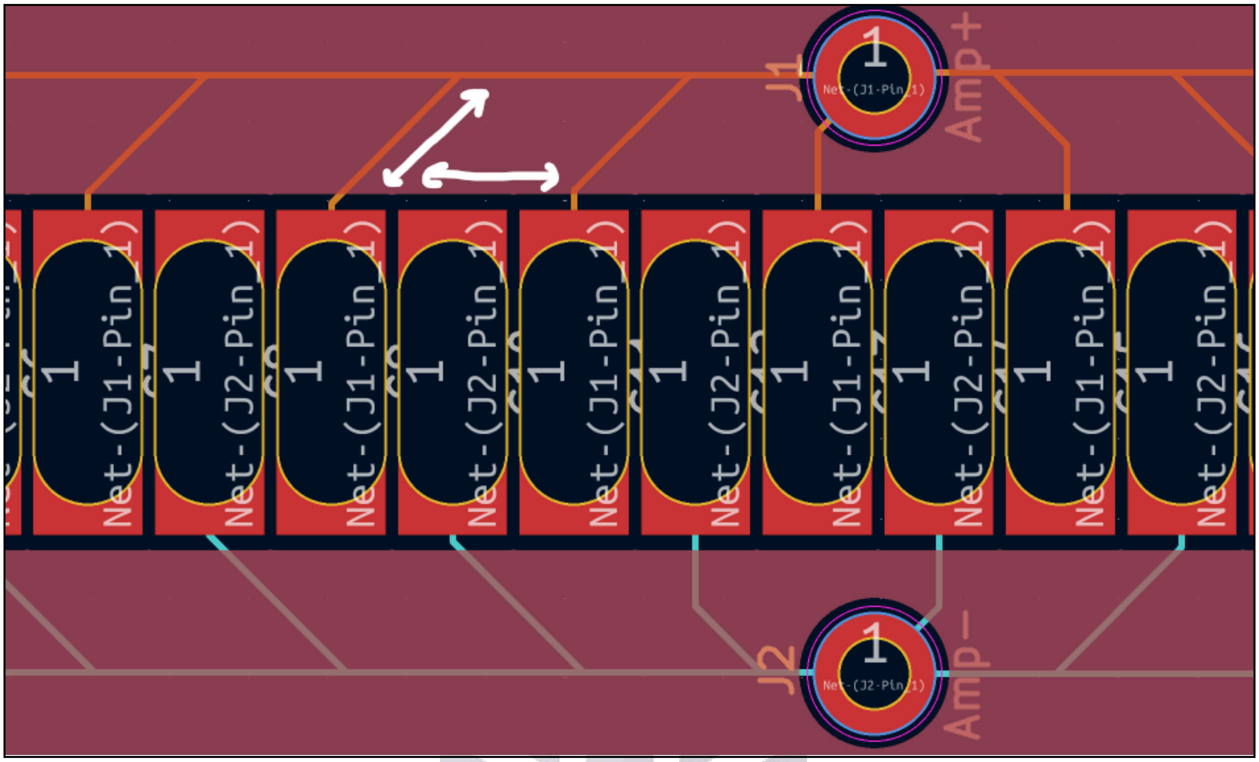


4.3 Features

4.3.1 Model 1:



The PCB utilizes a six-layer stackup in which the top and bottom layers serve as continuous ground planes, while the signal layers are embedded internally and separated from the ground planes by dielectric-only layers. This increased separation between the signal traces and reference planes reduces parasitic capacitance by increasing the effective dielectric spacing, thereby minimizing capacitive coupling. As a result, inter-trace crosstalk is reduced due to weaker electric field interaction between adjacent conductors. Additionally, the reduced coupling between the signal layers and their return paths lowers unintended capacitive loading, which helps preserve signal integrity, particularly for high-frequency components.



The traces are routed at slight angles as they exit the pads to rapidly increase spacing between adjacent conductors. This reduces near-field electric coupling in the high-density pad region, minimizing parasitic capacitance and crosstalk. Additionally, the angled routing shortens the length over which traces run in parallel, further limiting electromagnetic coupling. This geometry also provides a more gradual transition from pad to trace, reducing impedance discontinuities and improving overall signal integrity. Moreover, the holes on the PCB are M2 for mounting purposes.

4.3.2 Model 2:

This model has the board properties as the previous board. The right half of model 2, is the same as model 1. There are independent ground pours between the left and right to separate high voltage from low voltage. On the high voltage side, it contains the accelerator. There are polygon borders to accommodate the high voltage and size 2512 resistors to safely step down the input voltage to each accelerator plate.

5. Testing and Future Improvements

5.1 Transmittence Testing

- Transmittence = area of holes / total surface area

5.2 Future Improvements

- Powerbed for more refined electrodes and further precision alignment
- Use oxygen free copper instead of stainless steel for electrodes
- Add amplifier to board